MPI LANCHESTER–OSIPOV

Verifiable Force

Quantitative assessment of the military power of states based on the Lanchester–Osipov model, with laws of combat measured on historical data and open validation of forecasts

Document 9 · Dissertation abstract

Proposed academic specialities: 1.2.2 “Mathematical modelling, numerical methods and software systems”; 2.3.1 “Systems analysis, control and information processing, statistics”; 5.5.4 “International relations, global and regional studies”

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For the executive and the investor: a one-page summary

What has been done. An open military power index of states (MPI) and a software suite for it have been created. It comprises a rating of 40 states across 253 indicators with uncertainty intervals, a combat model, a coalition builder, a calculator of missile and drone strikes against air defence, and a flashpoint monitor. The main difference from the market: **every claim of the model has been tested against real battles and wars under a protocol fixed before computation, and misses are published**.

Question	Answer
What task it solves	Comparison of the military power of states and coalitions for a specific theatre and horizon, with an honest measure of uncertainty rather than a single rating figure
Why now	World military expenditure in 2025 — 2.887 trillion $ (a SIPRI record), the wars of 2022–2026 have refuted counting ratings; demand for testable models is growing, while the niche leader TENET has moved from an index to an AI platform for analysts
How it is better than competitors	Open formulas and code; combat laws measured on 404 battles; validation on 10 wars of 2018–2025; explicit model error; a hot-spot monitor with AUC 0.96 out of sample (a simple baseline from border, history and last year gives 0.958 — we acknowledge this); competitors (Global Firepower, TENET MCI) disclose neither formulas nor validation
What already works	A Python calculation core (more than 60 automated tests, a reference ranking test; the code is closed until version 1.0, available to reviewers on request), a browser interface prototype, 9 Word reports, data collection from 30+ open sources, the Delphi expert panel protocol
What is missing so far	Coverage — 40 states against 145–170 for competitors; IISS data are not licensed; 67 “quality” indicators are temporarily filled with an AI estimate pending the expert panel; there is no sales team and no clients
Revenue model (options)	Platform subscription; a licence for the data and the methodology; commissioned scenarios and reports; training (military academies, think tanks)
The main risk	The model so far predicts war outcomes no better than a uniform forecast — this is acknowledged and published; the value of the product lies in an honest balance of power and in scenarios, not in “predicting wars”

> **Position.** unlike the market, we do not promise accurate forecasts of war outcomes: our own validation has shown that open data do not permit this. We sell testability — what competitors promise but keep closed.

1. General description of the work

Relevance of the topic

Assessment of the military power of states is needed for military planning, foreign policy, the defence industry, insurance and investors. Widely used ratings (Global Firepower, CINC) add up the quantity of equipment and resources linearly, with opaque or arbitrary weights, without accounting for quality, readiness, geography, time, allies or uncertainty. The wars of 2022–2026 showed the cost of these errors: in the 2022 GFP rating Russia stood in 2nd place and Ukraine in 22nd, yet the war became a protracted positional one; Israel and Iran stood close to each other in 2025, yet Israeli aviation achieved air supremacy over Iran within days.

The classical Lanchester–Osipov model (1915–1916) sets the form of the relationship between numbers and quality, but its laws are poorly supported by history, and when transferred to the level of states it loses economics, geography, alliances and the nuclear factor. A methodology is needed that links the index and the combat model and **tests itself against data**.

Degree of elaboration of the problem

Combat models evolved from the laws of Lanchester and Osipov to the modifications of Helmbold (1965) and Deitchman (1962), the Hughes salvo model (1995), three-sided models (Kress et al., 2018) and spatial models (Randow et al., 2026). Fitting to battles (Engel, 1954; Bracken, 1995; Fricker, 1998; Lucas, Turkes, 2004) produced contradictory results. Power indices (CINC, Souva MMP, DOE Carroll–Kenkel, Beckley) developed separately from combat models. Commercial ratings (Global Firepower, TENET MCI) are closed and do not publish validations. The chain “index → combat model → out-of-sample validation under a fixed protocol” is absent from the literature reviewed.

Aim and objectives

Aim — to create an open, testable methodology and software suite for the quantitative assessment of the military power of states and coalitions, linking the structural index with the combat model.

- To analyse combat models and power indices and identify their systematic errors.
- To build an index of 253 indicators with roles in the model, non-linear aggregation, a nuclear axis, power projection and uncertainty intervals.
- To measure the laws of combat (law exponent, intensity, quality factors, will threshold) on historical battles.
- To validate the model on the wars of 2018–2025 using data from the year before the war and to estimate its error.
- To develop models of coalitions, drone and missile strikes against air defence, and a flashpoint monitor.
- Create a software suite with automated tests and an interface; the methodology, formulas and parameters are open, the core code is closed until version 1.0.

Object and subject of research

Object of research — the military power of states and coalitions. **Subject of research** — mathematical models and methods for its quantitative assessment and their validation against historical data.

Scientific novelty

- **The logarithmic form of Lanchester's law as the rule for index weights:** $\ln F = \ln Q + \alpha \cdot \ln N$, where α is measured on battles rather than taken from a textbook; each of the 253 indicators is assigned a role in the combat equations.
- **Measuring combat laws on 404 battles of 1900–1973 (CDB90) under protocols fixed before the calculation:** the pre-registered $\alpha = 0.65$ [0.46; 0.84] rejects both the square and the linear law, and the pre-registered acceptance rule “estimate within [1; 2]” is not met; the exploratory effectiveness adjustment gives 0.98 [0.80; 1.18], but it lies outside the protocol and is not used for the conclusion; the form $p + q = 1$ is rejected for combat intensity ($p + q = 0.62$ [0.55; 0.70]).
- **Quality factors in units of numerical advantage:** one step of superiority in leadership equals an advantage of ≈ 9 times, in morale ≈ 6, in supply ≈ 2.3, in intelligence ≈ 2.2 (upper estimates: the codes were assigned after the events).
- **The will threshold with censoring:** the median losses at which a side stops fighting is 16% [14; 18]; one step of morale raises the threshold by a factor of 1.29. An independent survey (Podobnik et al., 2025) gives a median acceptable loss of 10%.
- **The defence coefficient of a specific war based on its loss ratio** (Russia against Ukraine — 1.7 according to Oryx, 2024–2025) and **an explicit model error $\sigma = 2.0$**, calibrated on 10 wars.
- **The first, as far as is known, out-of-sample test of the hypothesis that “the depth of production decides the drone war”** — the hypothesis is not supported.
- **An open log of negative results:** calibration, year-by-year quality, the strike model, drones — everything that did not improve the forecast is published with the reasons.

Theoretical and practical significance

Theoretically, the work clarifies the scope of Lanchester's laws: the pre-registered estimate rejects both classical laws on CDB90, while the exploratory effectiveness adjustment gives a nearly linear result (outside the protocol); combat intensity grows more slowly than numbers. The work provides measurable quality multipliers. Practically — a software package for comparing states and coalitions, war scenarios, air-defence assessment and hot-spot monitoring; the methodology is applicable in military academies, analytical centres, the defence industry and political-risk insurance.

Methodology and methods

Systems analysis; differential equations of combat; CES function; Monte Carlo method; regression and censored models (log-normal threshold model); evaluation of probabilistic forecasts by the logarithmic score (Gneiting, Raftery, 2007); leave-one-out cross-validation; Sobol sensitivity indices; Delphi method with Cooke weights; logistic regression for conflict forecasting; open-source data processing. **Working rule:** the protocol of every validation is fixed by a git commit before computation; the protocol cannot be changed after computation.

Propositions submitted for defence

- The military power index, in which indicators enter the formulas according to their roles in the combat equations and the pillars are aggregated with a CES function, eliminates the systematic errors of counting ratings: compensation of the “weakest link”, summing nuclear weapons with conventional forces, ignoring distance and uncertainty.
- On the battles of 1900–1973 the pre-registered estimate $\alpha = 0.65$ rejects both classical laws, and the acceptance rule [1; 2] is not met; the exploratory effectiveness adjustment gives a nearly linear result ($\alpha \approx 0.97$), but it is outside the protocol and not used for the conclusion; combat intensity grows more slowly than numbers ($p + q \approx 0.62$).

- The quality of troops (leadership, morale, supply, intelligence) is equivalent in strength to a several-fold numerical advantage and must enter the index as multipliers, not as addends.
- A battle ends with a collapse of will at median losses of about 16%, not with annihilation; the threshold is a scenario parameter (battle — 16%, campaign — 40%).
- The balance of power by the index from open data does not predict war outcomes better than a uniform forecast; an honest model must include an explicit error ($\sigma = 2.0$) and publish this.
- The probability of armed conflict in a pair of states is determined by proximity, history and institutions (AUC 0.96 out of sample; almost as much, 0.958, is given by a simple baseline from border, history and last year), not by the balance of military power.
- Open country-level features (GDP per capita, democracy, mechanisation) explain a fifth of the loss exchange in the battles of 1900–1973 but do not transfer to interstate clashes of 1989–2024 (UCDP GED: Spearman +0.25 with an interval from −0.60 to +0.63, 24 cases).

Reliability and approbation

Reliability is ensured by open formulas, parameters and protocols and by data with a hash manifest (the core code is closed until version 1.0, available to reviewers on request); more than 60 automated tests (checks of formulas, the battle model, the panel and calibration) run in GitHub Actions; a check of the browser calculation against the core calculation; protocols fixed before the calculation (commits 567224b, 65e1a59, 5ed015a, f2a1a3c, 67504ee, cb72541, 061ed46 and others); and publication of negative results. The materials are presented in 9 reports (documents 1–9) and an interactive prototype.

Structure of the work

The work consists of an introduction, eight chapters (review and criticism of models; the index; laws of combat on battles; validation against wars; drones and air defence; coalitions; the flashpoint monitor; the expert panel and the software suite), a conclusion, a bibliography and appendices (the catalogue of 253 indicators, validation protocols, data).

2. Main content of the work

Chapter 1. Combat models and power indices: a critical review

The Lanchester and Osipov laws (linear, quadratic, logarithmic), the modifications of Helmbold, Deitchman and Hughes, three-sided combat, and 13 indices and methods of power assessment are examined (Global Firepower, CINC, Lowy, BICC, Heritage, Souva, DOE, Beckley, Credit Suisse, Dupuy's QJM/TNDM, WEI/WUV and the “potential–antipotential” method, the domestic method of combat potentials). Common errors: counting “units” without quality and readiness, double-counting the budget and purchased equipment, summing nuclear weapons with conventional forces, linear aggregation, untested weights, point estimates without intervals, ignoring geography, time and allies.

Chapter 2. Military power index: construction

253 indicators are grouped into 12 pillars (ground forces, air force and air defence, navy, strategic forces, space and command, personnel, economy, defence industry, logistics, geography, alliances, governance and experience). Each indicator is assigned a role: N — numbers (41), Q — quality (57), M — multiplier (35), S — sustainment (81), P — projection (13), G — deterrence (11), C — context (15).

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(1) Normalisation of count quantities:  x̃ = lg(x) on a scale three orders of magnitude
below the sample leader
(2) Combat pillar (land, air, sea):
    P_d = (α·Σw·x̃[N] + Σw·x̃[Q,M] + Σw·x̃[other]) / (α·Σw[N] + Σw[Q,M] + Σw[other])
    — the logarithmic form of Lanchester's law  ln F = ln Q + α·ln N
(3) Pillar aggregation (CES):  MPI = (Σ ω_p · P_p^ρ)^(1/ρ),  ρ = 0.5
    ρ = 1 — sum; ρ → 0 — geometric mean; ρ < 0 — “weakest link”
(4) Nuclear axis: class 0–4 outside the aggregation; modes “axis”, “pillar”,
“multiplier” (1 + 0.3·P_STR)/1.3
(5) Power projection (Boulding):  MPI(D) = MPI · e^(–D/λ),  λ = λ₀ + λ₁·P̄[role P]
(6) Conversion into c.u. of power:  M = exp(κ·MPI/100),  κ = 2.6·ln 10 ≈ 6.0
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Uncertainty: each observation is assigned a reliability class (A–D) and an error σ; the Monte Carlo method (1,000 runs) varies the data, the weights (× LogN(0; 0.15²)), ρ ∈ [0.3; 0.7] and α ∈ [0.8; 1.2]; the result is the MPI interval and the range of ranks. Pilot: 40 states (20 major powers and 20 flashpoint participants), 215 indicators in the index, the snapshot is 83.9% complete (without the temporary AI estimate — 57.4%). Result: United States 79.6 [77.2; 80.0], China 75.5, Russia 70.4; the rank correlation with the expert assessment of 8 countries — 0.81. Sensitivity (Sobol): model structure — 0.36 of the spread, data — 0.34.

Chapter 3. Laws of combat on historical battles

The material is the US Army's CDB90 database: 404 battles of 1900–1973 with numbers, losses, duration and the compilers' expert factors. All calculations follow protocols fixed before computation.

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(7) Generalised combat model:  dA/dt = –a · A^(1–α/2) · B^(α/2) + r_A(t)
                               dB/dt = –b · B^(1–α/2) · A^(α/2) + r_B(t)
    invariant without reinforcements:  b·A^α – a·B^α = const
(8) Required force ratio:  R = (e_B · d / e_A)^(1/α)
(9) Combat intensity:  ln(L/T) = c + p·ln N_enemy + q·ln N_own
(10) Factor multiplier in units of advantage:  eq = exp(β_factor / β_ln(N_A/N_B))
(11) Will threshold (censored log-normal model):  ln m* ~ N(μ, s²),
     observed losses f — the lower bound of the threshold for the side that holds
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Result	Value	Conclusion
Law exponent α (under the protocol)	0.65 [0.46; 0.84]	the square law (α = 2) rejected
α adjusted for effectiveness	0.98 [0.80; 1.18]	exploratory estimate, outside the protocol; nearly linear law

Result	Value	Conclusion
Intensity: p, q, p + q	0.16; 0.46; 0.62 [0.55; 0.70]	doubling of both sides → daily losses × 1.54, not × 2: only part of the forces takes part in battle
Leadership / morale / supply / intelligence	≈ 9 / 6 / 2.3 / 2.2 times the advantage	quality matters more than numbers (upper estimates)
Attackers win	68% of battles	the force ratio hardly improves the forecast
Will threshold m*, median	16% [14; 18]	combat ends with a collapse of will
Morale step	threshold × 1.29 [1.12; 1.48]	morale raises staying power

Chapter 4. Validation against wars and model error

Calibration against 5 wars of 2020–2026 (1,350 parameter combinations, cross-validation): the parameters that are best in the sample are worse than the defaults on held-out wars (−31.7 versus −19.6) — the defaults were retained; there is no common κ at any α. **Extended validation against 10 wars of 2018–2025** using the UCDP register, with the index based on data from the year before the war (72 identical indicators): all model variants are worse than the uniform forecast (−14.6); variant θC was adopted (α = 1.0; d = 1.3 / 1.0 / 1.0). **Model error:** σ = 2.0 in the probability that the advantage is sufficient:

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(12) P(the advantage is sufficient) = Φ( ln(r ÷ R) / σ ),  σ = 2.0
(13) Logarithmic score of the forecast:  S = Σ ln p(observed outcome)
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Defence coefficient of the Russia–Ukraine war based on Oryx equipment losses in 2024–2025 — d = 1.7. The scenario “Russia against Ukraine” yields a ratio of 1.68 : 1 against the required 1.70 : 1 — a positional war, which matches reality (but is not an independent validation). **Year-by-year quality** (expenditure per service member, education) and **a separate model of strike campaigns** did not improve the forecast — the results are published.

Chapter 5. Drones, salvoes and air defence

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(14) Hughes salvo model:  ΔB = (a·A − y_B·B) / w_B,   ΔA = (b·B − y_A·A) / w_A
(15) Air-defence saturation:  leakage share = (1−f)·(1−p)^n + f·(1−p)^(n+1),  k̄ = n + f
     (integer interceptors, uniform distribution — optimal according to arXiv
2608.31143)
(16) Drone hypothesis:  ln r_t = c + b·x_t,  x_t = ln(Ukraine's output ÷ Russia's
output)
```

Testing the hypothesis “the depth of production decides the drone war” (Gogerty, 2026) under protocol 061ed46: the ratio of Russian and Ukrainian equipment losses by year — 3.20; 2.31; 2.41; 1.16. Between 2023 and 2024 Ukraine's relative output fell (x from 1.74 to 0.38), while the loss ratio did not change (b ≈ −0.03); the forecast for 2025 is 2.38 against an actual 1.16, and the direction is wrong. **The hypothesis is not supported.** The literature (RUSI, 2025; Defence Studies, 2026) points to better inputs: 20–40% of drones reach the target, and about 90% of the fleet is expended within a year.

Chapter 6. Coalitions

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(17) F_coal(t) = F_own + Σ_j ρ_j · s_j · F_j · e^(−D_j/λ_j) · g(t − τ_j)
```

ρ is ally reliability (commitment classes A–F: from a collective defence treaty to an informal axis), s is the force share in theatre, τ is the entry delay. Contribution types: troops, supply, intelligence, territory, finances. Coalition scenarios use “autonomous power” without the alliances pillar so that allies are not counted twice. Scenarios: Russia against Ukraine (1.68 : 1 against the required 1.70); Israel and the United States against Iran (6.84 : 1 against 1.30); the Baltics (the attacker's window of advantage — days 0–7, then NATO advantage).

Chapter 7. Flashpoint monitor

780 pairs from 40 states × 2016–2025, outcome — armed conflict according to UCDP, features of the previous year. The model “proximity, distance, history, conflict last year, democracy (V-Dem), alliance” — **AUC 0.96 out of sample** (5,460 forecasts, 38 conflicts; the simple baseline “border + history + conflict last year” gives 0.958, the full model’s gain of +1.28 is below the threshold of 2 — the acceptance rule is not met); onsets of new conflicts — AUC 0.75 (8 cases). Military power indicators do not improve the forecast (−68.6 versus −66.7) and are not included in the monitor. The 2026 forecast is published in the prototype.

Chapter 8. Expert panel and software suite

Delphi panel: 67 questions in 9 competence modules, 0–4 scales with observable anchors or numbers with an 80% interval; competence coefficient $K_c \geq 0.5$; exclusion of assessments of the expert's own country; consensus — interquartile range; agreement — Kendall's W; Cooke calibration on 12 seed questions. Until the panel concludes, 67 indicators are filled with a temporary AI estimate (lowest reliability class; the calculation without it is published alongside).

Software suite: data collection from 30+ open sources (Python), the calculation core `core/lanmil` (normalisation, imputation, index, Monte Carlo, battle model, coalitions, salvos, validation, sensitivity), more than 60 automated tests in GitHub Actions, a single parameter file for the core and the prototype with a self-check, an interactive prototype (ranking, profile, comparison, battle model, salvos and air defence, coalitions, hot-spot monitor, methodology, catalogue), Word and Excel report generators.

3. Conclusion

Main conclusions

- Counting ratings are systematically wrong; an index with indicator roles, non-linear aggregation, a nuclear axis, projection and intervals eliminates these errors by construction.
- Lanchester's square law is not supported by history: the pre-registered α = 0.65 rejects the linear law as well (the rule [1; 2] is not met); the exploratory effectiveness adjustment gives a nearly linear result (0.97), but outside the protocol; combat intensity grows more slowly than numbers.
- The quality of troops (leadership, morale, supply, intelligence) is equivalent in strength to a several-fold numerical advantage; a battle ends with a collapse of will at losses of about 16%.
- From open data, the balance of power does not predict war outcomes better than a uniform forecast; the model must carry an explicit error, and this must be stated to the client plainly.
- Wars are not started by the strongest: the probability of conflict is determined by geography, history and institutions.
- The simple link “drone output → loss-exchange ratio” is not confirmed; measurable inputs are needed: consumption, the share reaching the target, losses to electronic warfare (EW).

Limitations

Coverage — 40 states; no licensed IISS data; 67 quality indicators are temporarily filled with an AI estimate; there are few validation wars (5–10), which limits calibration; CDB90 factors were assigned after the events and yield upper estimates; loss data for ongoing wars are biased by the sources.

Prospects

- An open forecast log: the first entry — monthly losses of Russia and Ukraine for Q4 2026, fixed before the period begins.
- An expert Delphi panel instead of the AI estimate; expansion to 150+ states.
- An attrition model with drones on measurable inputs; layered air defence (missiles + interceptor drones).
- Adjustment of loss data for source bias (Radford et al., 2023); checking the war's d on WarSpotting.
- Platform (FastAPI, React), a public website in two languages, a data DOI and a preprint on the CDB90 battles.

4. Competitive environment in the United States

Below are the US organisations closest to the MPI in terms of task. Important players outside the US are listed separately. Data are from the organisations' websites and the press as of 13.10.2026.

Organisation	Product	Strengths	Weaknesses	How we differ
Global Firepower (Virginia, Utah)	a rating of 145 countries, 60+ factors	coverage, recognition, annual cadence	the formula is closed, there are no intervals or validation; in 2022 the counting gap between Russia and Ukraine did not materialise	open formulas, intervals, validation against wars
TENET Intelligence (Arlington)	MCI: 150+ countries, 200+ indicators; since 2026 — the SAGE platform	an advisory board (former NATO commander, former intelligence chiefs), client testimonials	the core is closed; the square law is untested; the forecast log is closed; 29 of 50 sitemap addresses — 404	the combat law is measured; validations and misses are open
Heritage Foundation	Index of U.S. Military Strength	authority, expertise	only the US armed forces, a qualitative scale	comparison of states and coalitions
Verisk (Maplecroft)	Predictive War Index (May 2026): the probability of war in a country over 12 months	a risk-modelling brand, clients — banks and insurers	machine learning without published validation; trained on 1995–2022	a flashpoint monitor with a published out-of-sample AUC of 0.96 and a protocol
Rhombus Power	Guardian: AI decision support and early warning (US Air Force)	access to the customer — the Pentagon	closedness, confined to the defence sector	an open methodology for the civilian and expert market
US academic projects (COW/CINC; Souva MMP; DOE Carroll–Kenkel)	open datasets and measures of power	peer review, long data series	no product, scenarios or interface	a product on a validated methodology

Outside the United States	Why it matters
IISS Military Balance+ (United Kingdom)	the main source of data on the armed forces; a licence is needed to expand coverage
Janes (United Kingdom)	databases of 40 thousand military units; the benchmark of the data market
GlobalMilitary.net — ATLAS Index (country not identified)	169–170 countries, published weights, a worked example and a sensitivity table — the rating closest to ours in openness, but without a combat model or validation against wars
VIEWS (Norway, Sweden), ACLED	open early-warning systems for violence; a benchmark for the flashpoint monitor
Hadean (United Kingdom)	AI combat simulations, a round of about $15 million (2025–2026), investor — Booz Allen Ventures

Conclusion for the investor. the market is divided: ratings are open but untested; defence analytics platforms are closed and expensive. The niche of an “open, testable index with a combat model” is vacant. Demand is confirmed by US Air Force requests (WarMatrix — scenarios 10,000 times faster than real time) and by insurers entering war modelling.

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